

Patricia Magumise

Predicting Movie Revenue Using Budget, Audience Engagement, Popularity, and Production Features

1. INTRODUCTION

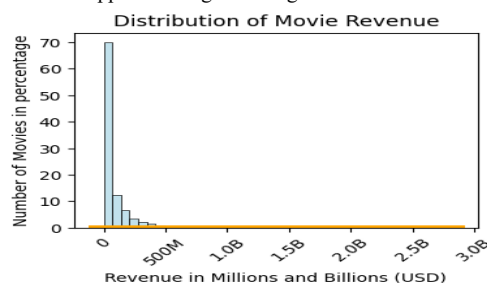
This project predicts movie revenue from production features such as budget, original language, popularity, runtime, and production-related attributes. Because revenue is a continuous numeric variable, this is a regression problem with revenue as the target. Accurate revenue prediction helps studios, distributors, and investors make better marketing, release-timing, and investment decisions. It is of utmost importance to understand the indicators that lead to the success of a movie. Predicting movie revenue can help companies and investors make good decisions concerning how they can market, plan, and make decisions.

The dataset interests me because it connects my background in information science with my current studies in data science. I'm especially interested in how metadata and categorical information can be used alongside numerical features for prediction and decision-making. Since movie data relies heavily on metadata such as genres, budget, and language, it is relevant to my experience and provides an opportunity to apply machine learning to real-world data.

The [TMDB movie dataset](#) was obtained from Kaggle and contains information such as budget, popularity, release date, status, average vote, runtime, language, and production-related features. The dataset falls under entertainment and commerce.

2. EXPLORATORY DATA ANALYSIS (EDA)

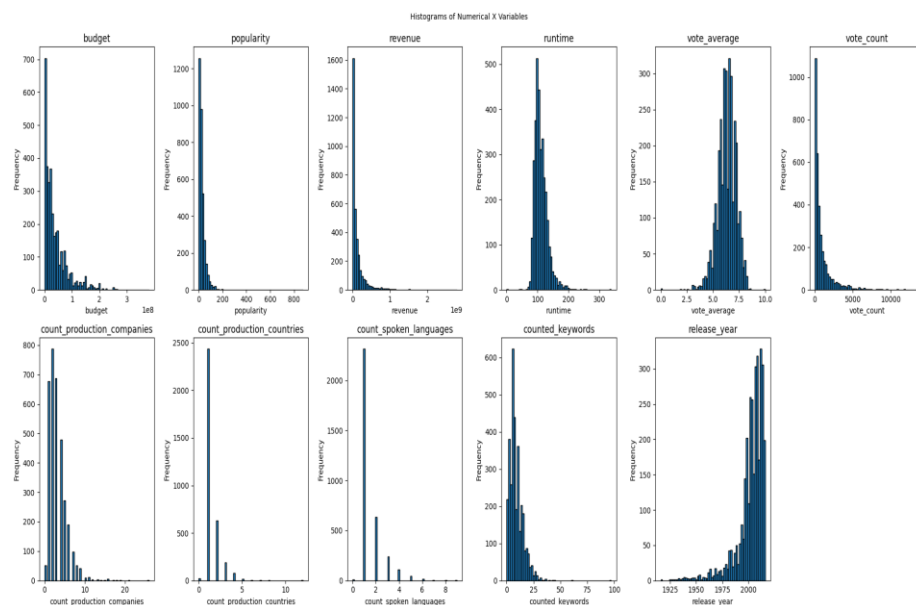
- The dataset shape is 3376 rows and 15 columns after preprocessing.
- Budget and revenue are reported in US dollars; vote average uses a 0–10 scale.
- Each row represents one movie and its associated characteristics.
- The target variable is highly right-skewed: about 49% of movies earn under \$50M, while fewer than 1% exceed \$1B. This non-normal distribution motivates the log-transform applied during modeling.



- Missing data was concentrated in homepage (3,091 missing, 64%) and tagline (844 missing, 18%), both of which were dropped as they offer no predictive value. The target variable had 1427 missing values which I dropped because revenue = 0 likely indicates unreported data not predictive. The genres clean feature had 3375 non-null entries after processing. Runtime had 2 missing values and release date had 1; both

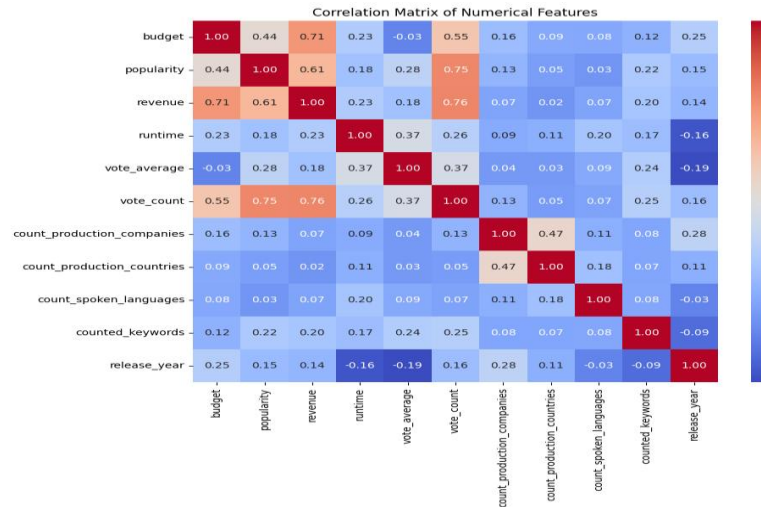
were handled via imputation in the preprocessing pipeline. All other features were complete with 3376 non-null observations.

- Key features includes budget (production spend), popularity (TMDB engagement score), vote average and vote count (audience ratings), runtime (film length), original language and status (categorical), genres clean (text), release month (engineered cyclic feature), and engineered counts for production companies, countries, spoken languages, and keywords.



- Budget, popularity, vote count, counted keywords, and the production-count features all show heavy right-skew as most movies cluster near zero, with a long tail of blockbusters.
- This skew is the main reason a linear model alone is unlikely to fit well as it inflates the influence of a small number of outliers and motivated both the log-transform on revenue and the choice to test tree-based models that handle non-linear, threshold-style splits.
- Runtime, in contrast, is roughly bell-shaped around 90-110 minutes which shows there is little variation, runtime is unlikely to drive large revenue differences on its own.
- Vote average is also bell-shaped around 6 with a hard floor of zero, the bounded scale and lack of spread mean on this feature carries less discriminative power than vote

count, which captures volume rather than sentiment.

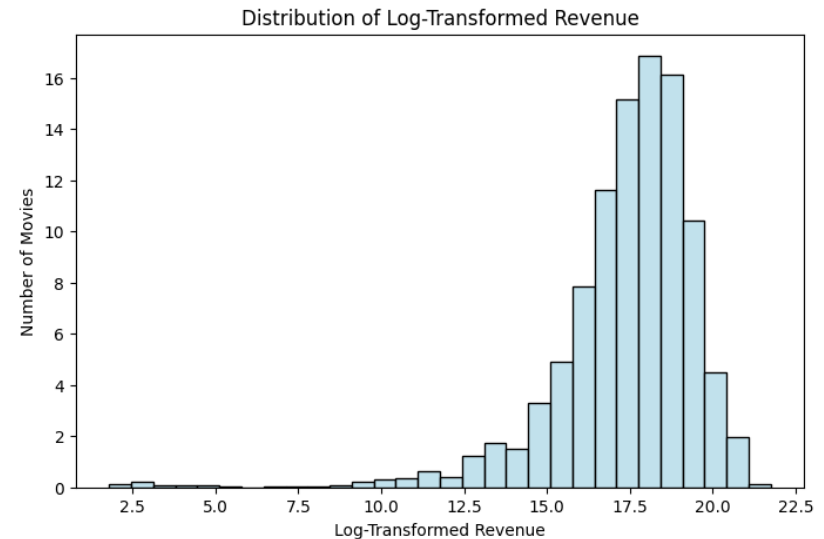


- Three features carry the bulk of the signal for revenue: vote count ($r = 0.76$), budget ($r = 0.71$), and popularity ($r = 0.61$). These three are the only correlations with the target above 0.5, which is consistent with the Random Forest importance results as the same three features dominate.
- Popularity and vote count are strongly correlated ($r = 0.75$), and budget overlaps moderately with vote count ($r = 0.55$) and popularity ($r = 0.44$), so there is real redundancy among the top predictors; this is why the linear Ridge model underperforms relative to the tree-based models, which can exploit non-linear interactions among them rather than treating them as independent inputs.
- The other notable relationship is between count production companies and count production countries ($r = 0.47$), which is expected since larger co-productions involve partners across borders.
- Every remaining correlation falls below 0.40 including vote average with revenue ($r = 0.18$), suggesting that how many people vote matters far more than how highly they vote.

3. DATA PREPARATION AND PREPROCESSING

- Nested JSON-like columns (genres, production companies, production countries, spoken languages, keywords) were parsed and converted to numeric count features, making them suitable for modeling. Duplicate rows were removed and irrelevant columns (id, title, homepage, overview, tagline, original title) were dropped as they carry no predictive signal. This simplified the data and made it suitable for modeling.

- The data was split into three subsets: 80% training, 10% validation, and 10% test (random state 42). These splits were fixed and used for all model fitting the release date variable was converted to datetime format and a new feature, release month, for cyclic transformations. First removed duplicates and missing values
- Imputation was performed in preprocessing pipeline.
- The target variable, revenue was log-transformed to reduce skewness and the impact of extreme values. Class balancing is not applicable since this is a regression problem.



- Yes I removed, id, title, homepage, overview, tagline, original title, as those do not contribute to a meaningful predictive analysis.
- The preprocessing pipeline routes numerical, categorical, text, and time features through dedicated branches, with TF-IDF used for the genres text feature.
- Categorical columns such as status and original language were encoded using one-hot encoding.
- text feature genres clean was included and transformed using a TF-IDF vectorizer.
- Yes I had time feature, the release month feature was engineered from the release date. Since months are cyclical in nature, a cyclic transformation using sine and cosine functions was used.
- The target variable, revenue showed right skewness, it was very uneven, with a small number of movies earning extremely high amounts. To solve this, a log transformation was used using a TransformedTargetRegressor to improve the model. After training, predictions were converted back to the original scale so they can be understood in terms of actual revenue.
- Scaling was applied using standard scaler, to ensure use of same scale.

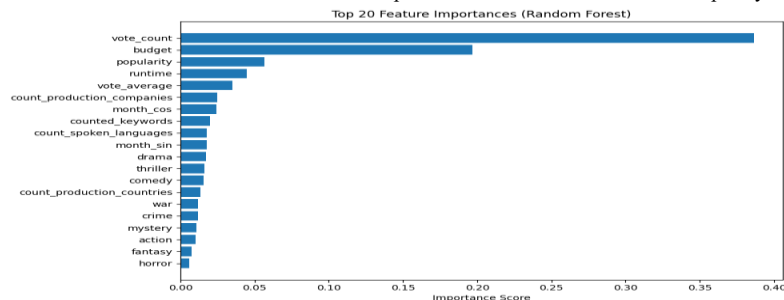
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- Missing values were handled inside the preprocessing pipeline using median imputation for numerical features and most-frequent imputation for categorical features. Class balancing is not applicable since this is a regression problem; the analogous skew in the target was addressed via the log-transform.
- Encoded categorical variables such as original language, status using one-hot encoding since there is no natural order.

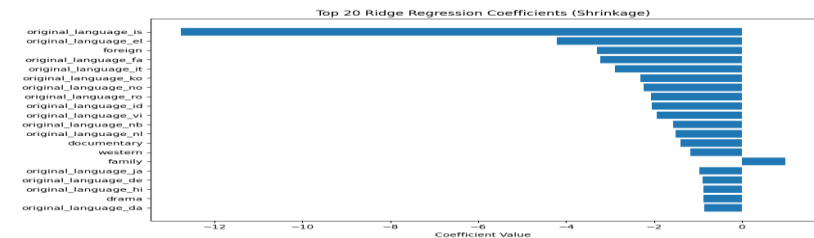
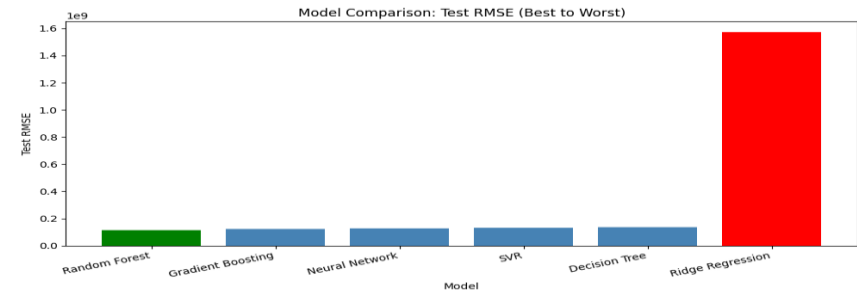
4. MODEL TRAINING AND RESULTS

	Model	Hyperparameters Tried	Best Hyperparameters	Mean CV RMSE	Standard Deviation	Test RMSE
1	Ridge Regression	alpha: [0.01, 0.1, 1, 10]	{'ridge_reg_regressor_alpha': 0.1}	281160645664.099976	559961971318.310059	1.572582e+09
2	Decision Tree	max_depth: [3, 5, 10, 15, None], min_samples_split: [2, 5, 10]	{'dt_reg_regressor_max_depth': 5, 'dt_reg_regressor_min_samples_split': 2}	101988571.05	5764843.18	1.367215e+08
3	SVR	C: [0.1, 1, 10, 100], kernel: ['rbf'], gamma: ['scale', 'auto'], gamma_max: ['scale', 'auto']	{'svr_reg_regressor_C': 1, 'svr_reg_regressor_gamma': 'auto', 'svr_reg_regressor_kernel': 'rbf'}	100152220.63	6476668.83	1.331820e+08
4	Random Forest	n_estimators: [100, 200], max_depth: [5, 10, None], min_samples_split: [2, 5]	{'rf_reg_regressor_max_depth': None, 'rf_reg_regressor_min_samples_split': 2, 'rf_reg_regressor_n_estimators': 200}	86422343.13	3827794.96	1.140373e+08
5	Gradient Boosting	learning_rate: 0.05, max_depth: 4, subsample: 0.8, n_iter_no_change: 20	Early Stopping (n_iter_no_change=20)	N/A	N/A	1.233898e+08
6	Neural Network	layers: [1,2,3], neurons: [10,50,64], activation: [relu,tanh], dropout: [True/False]	scenario_01 — {n_layers: 1, n_neurons: 10, activation: 'relu', use_dropout: False, optimizer: 'adam'}	N/A	N/A	1.217138e+08

RMSE was chosen as the primary metric because it is in the same units as the target (dollars) and penalizes large errors most heavily appropriate when blockbuster predictions matter. Random Forest achieved the lowest Test RMSE (\$114M) and the smallest CV standard deviation (\$3.8M), making it both the most accurate and most stable model. Decision Tree (\$137M) and SVR (\$133M) landed in the middle., as decision Tree underfits without an ensemble, and SVR's RBF kernel captures non-linearities but doesn't aggregate weak learners. Gradient Boosting (\$123M) and the Neural Network (\$122M) approach Random Forest's accuracy at higher complexity; their CV columns are N/A because the rubric specifies early stopping for GB and configuration testing for NN. Ridge's CV mean and std appear unusually large because TransformedTargetRegressor applies expm1 to predictions before scoring, so a small log-scale error becomes a billion-dollar error after exponentiation; Test RMSE (\$1.57B) is therefore the trustworthy comparison for Ridge and confirms a linear model cannot capture the non-linear structure. Ridge Regression is the simplest model (5/5) and remains competitive. Overall, Random Forest achieves the best balance between performance and simplicity.



Random Forest feature importance identified vote count, budget and popularity as the three strongest predictors of revenue. This aligns with domain intuition that higher-budget films receive more marketing and distribution, while popularity and votes reflect audience interest prior to wide release.



	Model	Mean CV RMSE	Std. Dev.	Test RMSE	Simplicity
1	Ridge Regression	281160645664.099976	559961971318.310059	1.572582e+09	5/5
2	Decision Tree	101988571.05	5764843.18	1.367215e+08	3/5
3	SVR	100152220.63	6476668.83	1.331820e+08	4/5
4	Random Forest	86422343.13	3827794.96	1.140373e+08	3/5
5	Gradient Boosting	N/A	N/A	1.233898e+08	2/5
6	Neural Network	N/A	N/A	1.268426e+08	1/5

Next steps and insights to improve model

- Gradient Boosting and Neural Network showed N/A for CV Mean/Std this gap should be addressed so all models are evaluated on equal footing.
- Adding richer features such as cast popularity, director experience, franchise membership, and release season are strong revenue predictors absent from the current dataset.